Shrinkage Helps, Nonlinearity Helps More: Linear and Neural Network Regression of the Aqueous Solubility of 1,267 Compounds

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***Abstract*—This work compares linear and nonlinear regression models for predicting the aqueous solubility of chemical compounds from molecular descriptors. The dataset comprises $1{,}267$ compounds described by 228 predictors, 208 binary structural fingerprints, 16 counts and 4 continuous descriptors, split into 951 training and 316 test observations. After a Yeo-Johnson transformation that reduces the median absolute skewness of the non-binary predictors from 1.36 to 0.06, we fit ordinary least squares (OLS), ridge regression and partial least squares (PLS), all implemented from scratch together with 10-fold cross-validation and checked against scikit-learn, and a single hidden layer neural network, comparing them through repeated cross-validation with corrected resampled t-tests. The three linear models are statistically indistinguishable on the test set, with RMSE between 0.717 and 0.743 and R^2 between 0.872 and 0.881, whereas the neural network reaches an RMSE of 0.620 and an R^2 of 0.911, a difference significant against every linear model, which indicates that part of the structure–solubility relationship is nonlinear or interactive.**

***Index Terms*—regression, ridge regression, partial least squares, neural networks, cross-validation, aqueous solubility, QSPR**

I. Introduction

Regression is the task of learning, from N observed pairs $(\mathbf{x}_n, y_n)$, a function f that predicts a continuous outcome y from a vector of D predictors $\mathbf{x}$ [1], [2]. The simplest and most interpretable choice is a linear function fitted by ordinary least squares (OLS), which is unbiased but becomes unstable when predictors are numerous and collinear, because the variance of its coefficients grows with the ill-conditioning of $\mathbf{X}^\top\mathbf{X}$.

Two families of remedies keep the model linear while trading a small bias for a large reduction in variance. Penalised regression shrinks the coefficients, either with an L_2 penalty (ridge regression [3]) or with an L_1 penalty that also selects variables (the lasso [4]). Latent variable methods instead regress the outcome on a few linear combinations of the predictors: principal component regression (PCR) builds them from the predictors alone, while partial least squares (PLS) builds them to covary with the outcome [5]. When the relationship is not linear, a multilayer perceptron with one hidden layer, trained by back-propagation [6], can approximate any continuous function given enough hidden units [7], at the price of a non-convex fit and of a much larger number of parameters.

A classical application of these models is quantitative structure–property relationship (QSPR) modelling in chemistry, and aqueous solubility is one of its most studied targets, since a poorly soluble compound is poorly absorbed and seldom becomes a drug. Solubility has been predicted from topological indices with linear and neural models [8], [9], and simple linear equations on a handful of descriptors such as ESOL remain a reference in the field [10]. The dataset used here is the benchmark of Kuhn and Johnson [2], derived from those earlier compilations.

This work fits and compares four regression models on that benchmark: OLS, ridge regression, PLS and a neural network. The linear models and the cross-validation are implemented from scratch and checked against scikit-learn [11], and the comparison asks not only which model has the lowest error but whether the differences are statistically significant. Section II describes the data and the methods, Section III reports and discusses the results, and Section IV concludes.

II. Methods

A. Dataset

The dataset contains $N = 1{,}267$ compounds, provided already split into $N_{tr} = 951$ training and $N_{ts} = 316$ test observations, and $D = 228$ predictors with no missing values: 208 binary fingerprints indicating the presence of a chemical substructure, 16 counts (atoms, bonds, rings, halogens, etc.) and 4 continuous descriptors (molecular weight, a hydrophilic factor and two surface areas) [2]. The outcome is the base-10 logarithm of the solubility, ranging from -11.62 to 1.58 with mean -2.72 on the training set and an almost identical

distribution on the test set (mean -2.80, standard deviation 2.08 against 2.05), so the provided split is representative. The fingerprints are sparse but not degenerate, with frequency of ones between 3.8% and 86%.

B. Pre-processing

All 20 non-binary predictors are right-skewed, with skewness between 0.67 and 3.84 (median 1.36). To remove it we apply the Yeo-Johnson transformation [12], a generalisation of the Box-Cox family [13] that accepts zeros, which are common among the counts, and negative values, present in the hydrophilic factor:

$$\psi(x,\lambda) = \begin{cases} [(x+1)^{\lambda}-1]/\lambda, & x \geq 0,\ \lambda \neq 0, \\ -[(1-x)^{2-\lambda}-1]/(2-\lambda), & x < 0,\ \lambda \neq 2, \end{cases} \tag{1}$$

with the logarithmic limits at $\lambda = 0$ and $\lambda = 2$. Each λ is estimated by maximum likelihood over $[-20, 20]$. Afterwards every predictor, including the fingerprints, is standardised to zero mean and unit variance. All parameters are estimated on the training set only and applied unchanged to the test set; inside cross-validation they are re-estimated in every fold from that fold's training part, so that the held-out part never informs the pre-processing.

TABLE I
YEO-JOHNSON PARAMETER AND SKEWNESS γ BEFORE AND AFTER THE TRANSFORMATION FOR SELECTED PREDICTORS (TRAINING SET)

Predictor	$\hat{\lambda}$	γ raw	γ transf.	Zeros
MolWeight	0.00	0.99	-0.00	0%
SurfaceArea1	0.35	1.71	-0.23	23%
HydrophilicFactor	-0.32	3.40	0.25	0%
NumCarbon	0.25	0.93	-0.00	0%
NumRotBonds	-0.08	1.58	0.02	29%
NumOxygen	-0.17	1.77	0.04	34%
NumChlorine	-3.63	3.18	1.42	79%
NumSulfer	-9.77	3.84	2.24	87%

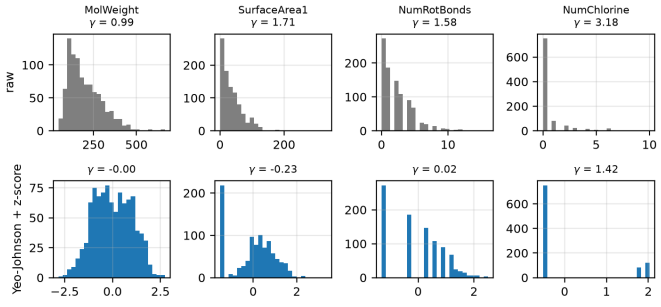


Fig. 1. Histograms of four predictors before (top) and after (bottom) the Yeo-Johnson transformation and standardisation.

Table I and Fig. 1 show the effect. The continuous descriptors and the counts with many distinct values become practically symmetric, and the median absolute skewness of the 20 non-binary predictors drops from 1.36 to 0.06. The exceptions are the counts dominated by zeros: sulfur, chlorine and halogen counts are zero in 72% to 87% of the compounds and keep a skewness between 1.0 and 2.2, because no monotonic transformation can spread a point mass, only compress the tail (bottom right of Fig. 1).

C. Correlation structure

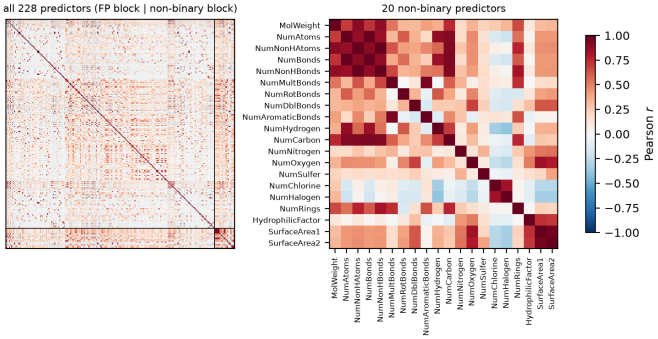


Fig. 2. Pearson correlation matrix of the transformed training predictors: all 228 (left) and the 20 non-binary ones (right).

Fig. 2 shows the correlation matrix of the transformed predictors. Among the $25{,}878$ pairs, the median absolute correlation is only 0.091, but 47 pairs exceed 0.9 and 7 exceed 0.99. The strongest dependences are of two kinds. Size descriptors measure the same property in different ways: number of atoms and of bonds ($r = 0.997$), of non-hydrogen atoms and bonds (0.995), and molecular weight, carbon count and surface areas forming the dark block of the right panel. And some fingerprints are near copies of a count, such as FP134 and the sulfur count ($r = 1.000$) or FP107 and the chlorine count (0.999). As a consequence the eigenvalues of the correlation matrix range from 28.3 down to 2.9×10^{-8}, a condition number of 10^9, and 22 eigenvalues are below 0.01. This is the regime in which OLS coefficients are unstable and shrinkage or latent variable methods should help.

Fig. 3 relates each non-binary predictor to the outcome. The strongest individual correlations are all negative and all measure molecular size: molecular weight ($r = -0.66$), carbon count (-0.61) and non-hydrogen bonds and atoms (-0.58), which agrees with the chemical intuition that larger molecules are less soluble in water. Only six predictors have $|r| > 0.5$ and the median $|r|$ over all 228 is 0.096; the strongest fingerprint, FP076, reaches $r = -0.54$. To judge linearity quantitatively we compared, for each non-binary predictor, the R^2 of a univariate straight line with that of a cubic polynomial. The gain is below 0.05 for 12 of the 20 predictors, so after the transformation most individual relations are close to linear. The largest gains, up to 0.13, occur for the aromatic bond count, the surface areas and the halogen counts, all with a large share of zeros, where the red curves in Fig. 3 change slope between the zero and the positive values. For the fingerprints linearity holds trivially, since a binary predictor takes only two values.

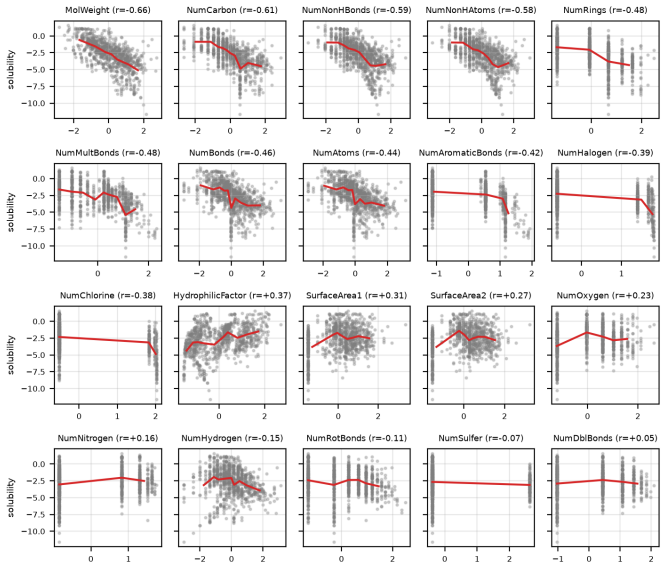


Fig. 3. Solubility against each transformed non-binary predictor, sorted by $|r|$, with the mean outcome in deciles of the predictor (red).

D. Regression models

OLS. With the design matrix $\mathbf{A} = [\mathbf{1}\ \mathbf{Z}]$ of the transformed predictors, the coefficients solve the normal equations

$$\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (\mathbf{A}^{\top}\mathbf{A})^{-1}\mathbf{A}^{\top}\mathbf{y}. \tag{2}$$

We solve the linear system directly and, when $\mathbf{A}$ is rank deficient, fall back to the minimum-norm solution $\mathbf{A}^{+}\mathbf{y}$ given by the Moore-Penrose pseudo-inverse. OLS is unbiased, has no tuning parameter and is fully interpretable, but with 229 coefficients and the collinearity above its variance is high.

Ridge regression adds an L_2 penalty on the slopes, leaving the intercept free. With centred $\mathbf{Z}_c$ and $\mathbf{y}_c$,

$$\hat{\boldsymbol{\beta}}_{\lambda} = (\mathbf{Z}_c^{\top}\mathbf{Z}_c + \lambda \mathbf{I})^{-1}\mathbf{Z}_c^{\top}\mathbf{y}_c. \tag{3}$$

Adding λ to every eigenvalue of $\mathbf{Z}_c^{\top}\mathbf{Z}_c$ removes the ill-conditioning and shrinks the coefficients toward zero [3]. Its cost is a bias that grows with λ, which must be tuned, and, unlike the lasso, it never sets a coefficient exactly to zero.

PLS regresses y on $A \ll D$ latent scores $\mathbf{t}_a = \mathbf{E}_{a-1}\mathbf{w}_a$, where the weight vector $\mathbf{w}_a \propto \mathbf{E}_{a-1}^{\top}\mathbf{f}_{a-1}$ maximises the covariance between the deflated predictors $\mathbf{E}$ and the deflated outcome $\mathbf{f}$ [5]. We implemented the NIPALS algorithm for a single outcome, whose coefficients in the original space are $\boldsymbol{\beta} = \mathbf{W}(\mathbf{P}^{\top}\mathbf{W})^{-1}\mathbf{q}$, with $\mathbf{P}$ the predictor loadings and $\mathbf{q}$ the outcome loadings. PLS was preferred to PCR because its components are guided by the outcome, so it usually needs fewer of them; its drawbacks are the tuning of A and components that are harder to interpret than principal components.

Neural network. We use a multilayer perceptron with one hidden layer of H hyperbolic tangent units and a linear output,

$$\hat{y} = b + \sum_{h=1}^{H} v_h \tanh\!\left(c_h + \mathbf{u}_h^{\top}\mathbf{z}\right), \tag{4}$$

trained with L-BFGS on the squared error plus a weight decay $\alpha\|\mathbf{w}\|^2$, using scikit-learn [11], with the outcome standardised during training. The network can represent curvature and interactions that the linear models cannot [7], but it has $H(D+2)+1$ parameters, over 9,000 for $H = 40$, its fit is non-convex and depends on the random initialisation, and it is far less interpretable.

E. Model assessment

Performance is measured by $\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_n (y_n - \hat{y}_n)^2}$ and by $R^2 = 1 - \sum_n (y_n - \hat{y}_n)^2 / \sum_n (y_n - \bar{y})^2$. The K-fold cross-validation (CV) was implemented from scratch: the training indices are shuffled with a fixed seed and split into $K = 10$ folds, and every model and hyper-parameter is evaluated on the same partitions. Hyper-parameters are chosen by the minimum mean CV RMSE using the training set only: λ among 30 values log-spaced in $[10^{-2}, 10^{4}]$, A from 1 to 40, and (H, α) in $\{5, 10, 20, 40\} \times \{1, 3, 10, 30, 100\}$. The test set is used once, for the final models, with 95% bootstrap confidence intervals for the RMSE (2,000 resamples).

To decide whether two models differ, we repeat the 10-fold CV five times with new partitions and fixed hyper-parameters and apply to the 50 paired differences of RMSE the corrected resampled t-test of Nadeau and Bengio [14],

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{J} + \frac{N_{va}}{N_{tr}}\right) s_d^2}}, \tag{5}$$

with $J = 50$ and $J-1$ degrees of freedom; the term N_{va}/N_{tr} corrects the optimism of the ordinary paired t-test, whose samples are not independent because the training sets overlap. On the test set we apply the Wilcoxon signed-rank test [15] to the paired squared errors of each compound. In both cases the p-values of the six pairs are adjusted with Holm's procedure [16].

III. Results and Discussion

A. Ordinary least squares

TABLE II
OLS: IMPLEMENTATION FROM SCRATCH AGAINST SCIKIT-LEARN, AND TEST SET AGAINST CROSS-VALIDATION ESTIMATES (MEAN ± STANDARD DEVIATION OVER FOLDS)

Estimate	RMSE	R^2
Test set, from scratch	0.7430	0.8718
Test set, scikit-learn	0.7430	0.8718
10-fold CV, from scratch	0.7279 ± 0.068	0.8654 ± 0.040
10-fold CV, scikit-learn	0.7279 ± 0.068	0.8654 ± 0.040
5-fold CV, from scratch	0.7705 ± 0.060	0.8513 ± 0.046

Table II summarises the OLS model. The implementation from scratch and scikit-learn give the same test RMSE of 0.743 and R^2 of 0.872, with predictions equal to 10^{-8}; the coefficients agree only to 5×10^{-6}, which is what a condition number of 10^{9} allows. The two cross-validations agree fold by fold to 10^{-7}. The full training matrix has full rank, but

two of the ten folds do not: when part of the compounds is held out, some rare fingerprints become exact copies of another predictor, and there the pseudo-inverse branch of (2) is required, a failure mode that a naive solution of the normal equations turns into predictions of order 10^{12}. The training RMSE of 0.49 against a test RMSE of 0.74 shows the overfitting expected from 229 coefficients. The 10-fold estimate, 0.728 ± 0.068, slightly underestimates the test error, and the 5-fold estimate, 0.771 ± 0.060, slightly overestimates it, as expected since each 5-fold model is trained on only 80% of the data. The test RMSE lies between the two and within one standard deviation of both; with only 316 test compounds it carries itself an uncertainty of about ± 0.08 (Table III), so CV and test set are consistent.

B. Ridge regression

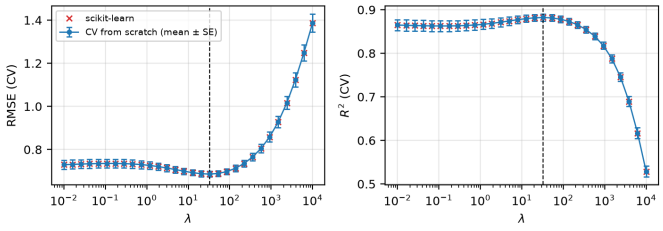


Fig. 4. Ridge: 10-fold CV profile of RMSE (left) and R^2 (right) against λ, mean $\pm$ standard error over folds; red crosses are scikit-learn's `GridSearchCV`. The dashed line marks the selected λ.

Fig. 4 shows the CV profile. For small λ ridge coincides with OLS; the RMSE then decreases to a minimum of 0.686 at $\lambda = 32.9$, where the CV R^2 is 0.882, and grows quickly once the penalty shrinks the coefficients too much, reaching 1.39 at $\lambda = 10^4$. The profile computed by scikit-learn on the same folds overlaps ours to within 10^{-8} and selects the same λ, and for a fixed λ the two coefficient vectors agree to 5×10^{-14}. At the selected value the norm of the coefficient vector drops from 152 (OLS) to 1.5, and the test RMSE improves from 0.743 to 0.717 ($R^2 = 0.881$). The one-standard-error rule would choose the more regularised $\lambda = 85$ at almost the same CV error.

C. Partial least squares

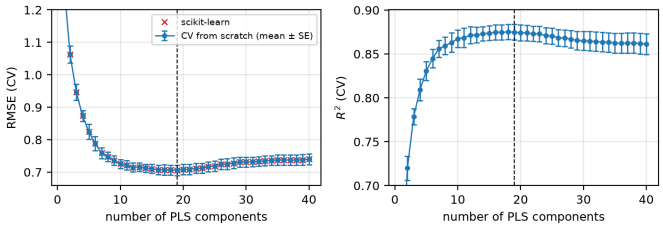


Fig. 5. PLS: 10-fold CV profile of RMSE (left) and R^2 (right) against the number of components; red crosses are scikit-learn's `PLSRegression`. The dashed line marks the selected value.

Our NIPALS implementation matches scikit-learn's `PLSRegression` to 2×10^{-15} in the coefficients. Fig. 5 shows that the CV RMSE falls steeply over the first ten components, from 1.29 with one component to 0.73 with ten, reaches its minimum of 0.706 ($R^2 = 0.875$) at $A = 19$ and then slowly rises toward the OLS value as more components are added. With fewer than a tenth of the 228 dimensions PLS already matches OLS, confirming that the information on solubility is concentrated in a few latent directions. On the test set the 19-component model reaches an RMSE of 0.734 and an R^2 of 0.875. The one-standard-error rule would select only 11 components.

D. Neural network

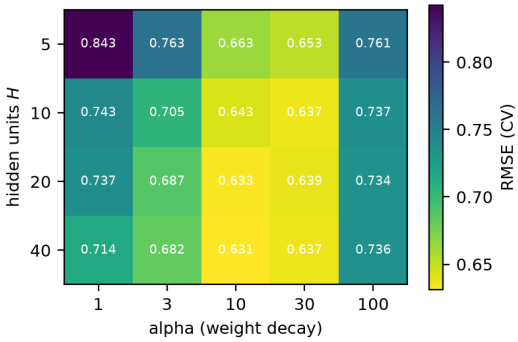


Fig. 6. Neural network: 10-fold CV RMSE over the grid of hidden units H and weight decay α.

Fig. 6 shows the CV RMSE over the hyper-parameter grid. Weight decay is decisive: with $\alpha = 1$ even large networks overfit (CV RMSE above 0.70), whereas α between 10 and 30 gives the best results for every H, and $\alpha = 100$ underfits. With adequate regularisation the error decreases with H and flattens beyond $H = 20$, the best configuration being $H = 40$, $\alpha = 10$, with a CV RMSE of 0.631 and R^2 of 0.900. On the test set it reaches an RMSE of 0.620 and an R^2 of 0.911. Retraining with ten random initialisations gives a test RMSE of 0.631 ± 0.010, so the result does not depend on a fortunate seed.

E. Model comparison

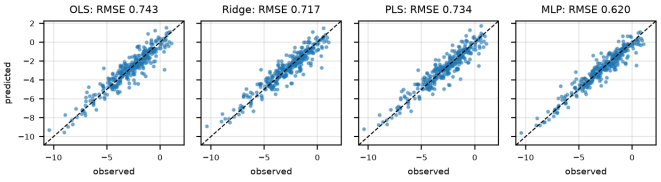


Fig. 7. Observed against predicted solubility on the test set.

Table III collects the results. The three linear models form a tight group, with test RMSE from 0.717 to 0.743 and R^2 from 0.872 to 0.881, and their confidence intervals overlap almost entirely. The neural network stands apart with a test RMSE of 0.620, 14% to 17% lower, and an R^2 of 0.911. Fig. 7 shows that all four models share the same pattern of shrinkage

TABLE III
PERFORMANCE OF THE FOUR MODELS: 10-FOLD CV ON THE TRAINING SET AND TEST SET, WITH 95% BOOTSTRAP CONFIDENCE INTERVAL FOR THE TEST RMSE

Model	Hyper-parameter	RMSE (CV)	R^2 (CV)	RMSE (test)	95% CI	R^2 (test)
OLS	—	0.728	0.865	0.743	0.668–0.822	0.872
Ridge	$\lambda = 32.9$	0.686	0.882	0.717	0.646–0.791	0.881
PLS	$A = 19$	0.706	0.875	0.734	0.663–0.806	0.875
MLP	$H = 40$, $\alpha = 10$	0.631	0.900	0.620	0.549–0.694	0.911

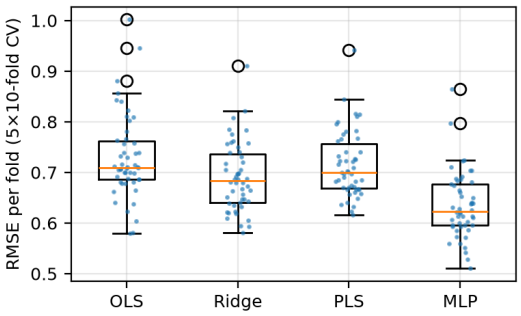


Fig. 8. RMSE per fold over five repetitions of 10-fold CV, with the hyper-parameters fixed at the selected values.

TABLE IV
PAIRWISE COMPARISONS: MEAN RMSE DIFFERENCE (FIRST MINUS SECOND MODEL) AND HOLM-ADJUSTED p-VALUES OF THE CORRECTED RESAMPLED t-TEST (REPEATED CV) AND OF THE WILCOXON TEST (TEST SET)

	Repeated CV		Test set	
Pair	ΔRMSE	p	ΔRMSE	p
OLS vs. Ridge	+0.039	0.081	+0.026	0.872
OLS vs. PLS	+0.014	0.386	+0.009	0.872
OLS vs. MLP	+0.095	< 0.001	+0.123	< 0.001
Ridge vs. PLS	-0.024	0.008	-0.016	0.205
Ridge vs. MLP	+0.056	< 0.001	+0.097	< 0.001
PLS vs. MLP	+0.080	< 0.001	+0.114	< 0.001

toward the mean: splitting the test compounds by solubility, the mean error (observed minus predicted) is between -0.19 and -0.43 for the 29 compounds below -6, which are predicted too soluble, and between $+0.53$ and $+0.60$ for the 20 above 0. The network does not remove this pattern, but it has the lowest RMSE in every one of the five ranges of solubility examined, so its gain is spread over the whole outcome range rather than confined to a few extreme compounds.

Fig. 8 and Table IV answer whether the differences are statistically significant. Among the linear models, neither OLS against ridge ($p = 0.081$ in CV) nor OLS against PLS is significant on either test, and ridge against PLS is significant in the repeated CV ($p = 0.008$) but not on the test set ($p = 0.21$): the advantage of ridge is consistent across folds but too small, 0.02 in RMSE, to be detected on 316 compounds. The neural network, in contrast, is better than every linear model with Holm-adjusted p-values below 0.001 on both tests. The standard deviation of the RMSE between folds, 0.07 to 0.08, is larger than the differences among the linear models, which is why a single split is not enough to rank them.

The nonlinear model therefore outperforms the linear ones, and the improvement is not an artefact of a particular partition or seed. This says that the relationship between the descriptors and solubility is not fully captured by an additive linear function, even after the transformation that made most individual relations nearly linear (Section II-C). Since the gain is spread over all ranges of the outcome, the remaining structure is most likely in interactions between predictors, such as the effect of a substructure depending on the size of the molecule, and in the residual curvature of the zero-inflated counts noted in Section II-C; a linear model can only represent these if they are engineered explicitly as new features. Shrinkage and latent components tame the variance of the linear fit, but they cannot add the missing nonlinear terms, which explains why ridge and PLS improve on OLS only marginally while the network improves on all of them.

IV. CONCLUSION

We fitted four regression models to the aqueous solubility of 1,267 compounds. The Yeo-Johnson transformation removed most of the skewness of the descriptors, and the exploratory analysis revealed strong collinearity among size descriptors and between fingerprints and counts, with a condition number of 10^9. The implementations from scratch of OLS, ridge regression, PLS and cross-validation reproduced scikit-learn to numerical precision, once rank-deficient folds were handled with the pseudo-inverse. Ridge and PLS controlled the variance caused by collinearity and improved slightly on OLS, but the three linear models remained statistically indistinguishable on the test set, with RMSE around 0.72–0.74. A single hidden layer neural network lowered the test RMSE to 0.62 and raised R^2 to 0.91, a significant improvement under both the corrected resampled t-test and the Wilcoxon test, which indicates that part of the structure–solubility relationship is nonlinear or interactive. Natural extensions are nonlinear models that remain interpretable, such as tree ensembles, and nested cross-validation, so that the hyper-parameter search is also accounted for in the comparison.

USE OF AI TOOLS

AI assistance (Anthropic Claude) was used in this work. The prompts, the responses and the use made of each are recorded in `USO_DE_IA.md` in the project repository.

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